

1. Instalacja Dockera

```
sudo apt update
sudo apt install docker.io -y
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
newgrp docker
```

Sprawdzenie działania:

```
kamilmarchewka@lima-MDO-server:~$ which docker
/usr/bin/docker
```

2. Zapoznanie się z obrazami

```
docker run hello-world
docker run busybox echo "Hello world!"
docker run ubuntu /bin/echo "Hello world!"
docker run fedora /bin/echo "hello"
docker run mariadb --version
docker run mcr.microsoft.com/dotnet/sdk:latest --version
docker run mcr.microsoft.com/dotnet/runtime:latest --help
docker run mcr.microsoft.com/dotnet/aspnet:latest --help
```

Sprawdzenie rozmiarów:

```
kamilmarchewka@lima-MDO-server:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	In Use
busybox:latest	1487d0af5f52	6.14MB	1.9MB	U	U
fedora:latest	781b7642e8bf	281MB	68.5MB	U	U
hello-world:latest	f9078146db2e	22.6kB	10.3kB	U	U
mariadb:latest	e0236fc6386e	492MB	109MB	U	U
mcr.microsoft.com/dotnet/aspnet:latest	55e37c7795bf	369MB	92.4MB	U	U
mcr.microsoft.com/dotnet/runtime:latest	8fb7ff015fcf	326MB	80.2MB	U	U
mcr.microsoft.com/dotnet/sdk:latest	8a90a473da52	1.31GB	321MB	U	U
ubuntu:latest	c4a8d5503dfb	141MB	30.8MB	U	U

```
kamilmarchewka@lima-MDO-server:~$
```

Najmniejszy rozmiar ma hello-world, natomiast najwięcej waży sdk

Sprawdzenie kodów wyjścia:

```
kamilmarchewka@lima-MDO-server:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
8668a8472185	mcr.microsoft.com/dotnet/aspnet:latest	"--help"	3 minutes ago	Created		kind_banach
f5e3d2030de5	mcr.microsoft.com/dotnet/runtime:latest	"--help"	4 minutes ago	Created		strange_wiles
612bea071276	mcr.microsoft.com/dotnet/sdk:latest	"--version"	4 minutes ago	Created		musling_babbage
435ede00d6b1	mariadb	"docker-entrypoint.s..."	5 minutes ago	Exited (0) 5 minutes ago		focused_hugle
527695fed77c	fedora	"/bin/echo hello\n\n"	6 minutes ago	Exited (0) 6 minutes ago		determined_meninsky
170143bcac33	ubuntu	"/bin/echo 'Pozdrowi..."	7 minutes ago	Exited (0) 7 minutes ago		elastic_cray
0edff71976c7	busybox	"sh"	8 minutes ago	Exited (0) 8 minutes ago		unruffled_solomon
de496289252d	busybox	"echo 'Hello world!'"	8 minutes ago	Exited (0) 8 minutes ago		elegant_goodall
0d2619704e69	hello-world	"/hello"	9 minutes ago	Exited (0) 9 minutes ago		festive_elbakyan

```
kamilmarchewka@lima-MDO-server:~$
```

3. Uruchomienie kontenera z obrazu busybox

```
[kamilmarchewka@lima-MDO-server:~$ docker run busybox echo "Hello Busybox"]  
Hello Busybox  
[kamilmarchewka@lima-MDO-server:~$
```

Podłączenie się do kontenera interaktywnie

```
docker run -it busybox sh
```

```
[kamilmarchewka@lima-MDO-server:~$ docker run -it busybox sh  
/ # vconfig  
BusyBox v1.37.0 (2024-09-26 21:31:42 UTC) multi-call binary.  
  
Usage: vconfig COMMAND [OPTIONS]  
  
Create and remove virtual ethernet devices  
  
    add IFACE VLAN_ID  
    rem VLAN_NAME  
    set_flag IFACE 0|1 VLAN_QOS  
    set_egress_map VLAN_NAME SKB_PRIO VLAN_QOS  
    set_ingress_map VLAN_NAME SKB_PRIO VLAN_QOS  
    set_name_type NAME_TYPE  
/ #
```

Numer wersji: v1.37.0

4. Uruchomienie systemu w kontenerze

```
[kamilmarchewka@lima-MDO-server:~$ docker run -it ubuntu bash  
[root@001f6598db11:/# ps -cf  
UID          PID     PPID  CLS PRI STIME TTY           TIME CMD  
root           1         0   TS   19 07:09 pts/0        00:00:00 bash  
root           9         1   TS   19 07:10 pts/0        00:00:00 ps -cf  
root@001f6598db11:/#
```

Id tego procesu na hoście

```
[kamilmarchewka@lima-MDO-server:~$ ps -ef | grep bash  
kamilma+ 196272    1743   0 08:32 pts/0    00:00:00 /bin/bash --login  
kamilma+ 196368  196272   0 08:34 pts/0    00:00:00 /bin/bash  
kamilma+ 197881  196368   0 09:09 pts/0    00:00:00 docker run -it ubuntu bash
```

Ten sam proces jest zarówno głównym procesem w kontenerze, jak i zwykłym procesem na hoście

Aktualizacja pakietów:

```
apt update
```

apt upgrade -y

```
1 package can be upgraded. Run 'apt list --upgradable' to see it.
[root@001f6598db11:/# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  libcap2
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 30.7 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 libcap2 arm64 1:2.66-5ubuntu2.4 [30.7 kB]
Fetched 30.7 kB in 1s (21.8 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
(Reading database ... 4376 files and directories currently installed.)
Preparing to unpack .../libcap2_1%3a2.66-5ubuntu2.4_arm64.deb ...
Unpacking libcap2:arm64 (1:2.66-5ubuntu2.4) over (1:2.66-5ubuntu2.2) ...
Setting up libcap2:arm64 (1:2.66-5ubuntu2.4) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
root@001f6598db11:/#
```

5. Dockerfile

```
Dockerfile
1 FROM ubuntu:22.04
2
3 RUN apt-get update && apt-get install -y \
4     git \
5     && rm -rf /var/lib/apt/lists/*
6
7 WORKDIR /app
8
9 RUN git clone https://github.com/Inzynieria0programowaniaAGH/MD02026 ITE.git .
10
11 CMD ["/bin/bash"]
```

Uruchomienie obrazu:

docker build -t obraz-z-git .

docker run -it obraz-z-git

```
root@7ba8f582168e:/app# ls -la
total 24
drwxr-xr-x 1 root root 4096 Apr 24 07:39 .
drwxr-xr-x 1 root root 4096 Apr 24 07:40 ..
drwxr-xr-x 8 root root 4096 Apr 24 07:39 .git
-rw-r--r-- 1 root root  45 Apr 24 07:39 README.md
drwxr-xr-x 2 root root 4096 Apr 24 07:39 READMEs
drwxr-xr-x 3 root root 4096 Apr 24 07:39 grupa4
```

Czyszczenie kontenerów:

docker ps -a

docker container prune -f

```
[kamilmarchewka@lima-MDO-server:~/MD02026_ITE/grupa4/KM419688$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
busybox:latest	1487d0af5f52	6.14MB	1.9MB	
fedora:latest	781b7642e8bf	281MB	68.5MB	
hello-world:latest	f9078146db2e	22.6kB	10.3kB	
mariadb:latest	e0236fc6386e	492MB	109MB	
mcr.microsoft.com/dotnet/aspnet:latest	55e37c7795bf	369MB	92.4MB	
mcr.microsoft.com/dotnet/runtime:latest	8fb7ff015fcf	326MB	80.2MB	
mcr.microsoft.com/dotnet/sdk:latest	8a90a473da52	1.31GB	321MB	
obraz-z-git:latest	6f70d96c24bc	1.08GB	486MB	
ubuntu:22.04	962f6cadeae0	109MB	29.6MB	
ubuntu:latest	c4a8d5503dfb	141MB	30.8MB	

```
[kamilmarchewka@lima-MDO-server:~/MD02026_ITE/grupa4/KM419688$ docker rmi obraz-z-git
Untagged: obraz-z-git:latest
Deleted: sha256:6f70d96c24bc6cba29d5751c5c56191ec212073cf9bfd6625a3b7cf99f8d8624
Deleted: sha256:77ad45906bdf0c2d29fcdd087d1ec3b9c48a093a5c0f8168d95720e4a77ce185
Deleted: sha256:5fee3ce8cccd9c8e722fd6f67aad98062f6eff60aa516b0672fd1f7bcbb0fd67
Deleted: sha256:16ca5fb44de186d4deadf4c28fcda25c3b72be221f1df9599faa6bd998cfb8d6
[kamilmarchewka@lima-MDO-server:~/MD02026_ITE/grupa4/KM419688$ docker images
```

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```

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```
[kamilmarchewka@lima-MDO-server:~/MD02026_ITE/grupa4/KM419688$ docker system df
```

TYPE	TOTAL	ACTIVE	SIZE	RECLAIMABLE
Images	0	0	0B	0B
Containers	0	0	0B	0B
Local Volumes	1	0	0B	0B
Build Cache	0	0	0B	0B

```
kamilmarchewka@lima-MDO-server:~/MD02026_ITE/grupa4/KM419688$
```